Math 5601: Introduction to Numerical Analysis
Homework assignment 3

Show all relevant work in detail to justify your conclusions. Partial credit depends upon the work you show.

Problem #1: Consider $f(t) = sin(t)$. (Simplify your solutions as much as possible)

(a) Without using the general inner product expression of the best approximation, find the best approximation $p(t) \in P^3[-1, 1]$ to $f(t) \in C[-1, 1]$ with respect to the L^2 norm.

(b) Find the Taylor polynomial approximation $p(t)$ of degree 3 at $t = 0$.

(c) Find the Lagrange polynomial approximation $p(t)$ of degree 3 that interpolates $f(t)$ at $t = -1, -\frac{1}{3}, \frac{1}{3}, 1$.

Problem #2: Based on $u_1 = 1$, $u_2 = x$, $u_3 = x^2$, use Gram-Schmidt orthogonalization process to compute the three polynomials $w_1(x)$, $w_2(x)$, and $w_3(x)$ which are orthonormal on the interval $[0, 1]$ with respect to the inner product $(f, g) = \int_0^1 f(x)g(x)\,dx$. Using these polynomials, find the best approximation in $P_2[0, 1]$ for $f(x) = x^{\frac{1}{2}}$.

Problem #3: Find the quadratic polynomial $p(t)$ that satisfies $p(0) = f(0)$, $p(2) = f(2)$, $p'(2) = f'(2)$.

Math 5601 - Homework 03 - Tyler Glidden

Problem 1

$f(t) = \sin(t)$

a.) W/out using general inner product, find the best approximation

$p(t) \in P^3[-1,1]$ with respect to L^2 Norm

$$p = \sum_{j=0}^{n} a_j u_j \Rightarrow \|f-p\| = \min \| f - \sum_{j=0}^{n} \gamma_j u_j \|$$

b.) Find the Taylor Polynomial of degree 3 at $t=0$

$$P_n(x)=\sum_{k=0}^{n}\frac{1}{k!}f^{(k)}(x_0)(x-x_0)^k$$

$f(t)=\sin(t)$ $\quad$ $f(0)=0$

$f'(t)=\cos(t)$ $\quad$ $f'(0)=1$

$f''(t)=-\sin(t)$ $\quad$ $f''(0)=0$

$f'''(t)=-\cos(t)$ $\quad$ $f'''(0)=-1$

$$f(0)+f'(0)t+\frac{1}{2!}f''(0)t^2+\frac{1}{3!}f'''(0)t^3$$

$$t-\frac{1}{6}t^3$$

c.) Find the Lagrange Polynomial $P(t)$ of degree 3 that interpolates $t=-1,-1/3,1/3,1$ ($f(t)$ for $t=x_0,x_1,x_2,x_3$)

$$P(x)=\sum_{k=0}^{n}f(x_k)L_k(x)$$

$$L_k(x)=\prod_{\substack{i=0\\ i\neq k}}^{n}\frac{x-x_i}{x_k-x_i}$$

$$L_0(x)=\frac{x-x_1}{x_0-x_1}\cdot\frac{x-x_2}{x_0-x_2}\cdot\frac{x-x_3}{x_0-x_3}=\frac{x+1/3}{-1+1/3}\cdot\frac{x-1/3}{-1-1/3}\cdot\frac{x-1}{-1-1}=\frac{3(x+1/3)}{3(-2/3)}\cdot\frac{3x-1/3}{-4/3}\cdot\frac{x-1}{-2}$$

$$=\frac{3x+1}{-2}\cdot\frac{3x-1}{-4}\cdot\frac{x-1}{-2}=\frac{9x^2-1}{8}\cdot\frac{x-1}{-2}=\frac{9x^3-9x^2-x+1}{-16}\ //$$

$$L_1(x)=\frac{x+1}{-1/3+1}\cdot\frac{x-1/3}{-1/3-1/3}\cdot\frac{x-1}{-1/3-1}=\frac{x+1}{+2/3}\cdot\frac{x-1/3}{-2/3}\cdot\frac{x-1}{-4/3}=\frac{(3x+3)}{+2}\frac{(3x-1)}{-2}\frac{(3x-3)}{-4}=\frac{(9x^2-3x+9x-3)(3x-3)}{16}=\frac{27x^3-9x^2-27x+9}{16}\ //$$

$$L_2(x)=\frac{x+1}{1/3+1}\cdot\frac{x+1/3}{1/3+1/3}\cdot\frac{x-1}{1/3-1}=\frac{x+1}{4/3}\cdot\frac{x+1/3}{2/3}\cdot\frac{x-1}{-2/3}=\frac{3x+3}{4}\cdot\frac{3x+1}{2}\cdot\frac{3x-3}{-2}=\frac{(9x^2-9)(3x+1)}{-16}=\frac{27x^3+9x^2-27x-9}{-16}\ //$$

$$L_3(x)=\frac{x+1}{1+1}\cdot\frac{x+1/3}{1+1/3}\cdot\frac{x-1/3}{1-1/3}=\frac{x+1}{2}\cdot\frac{x+1/3}{4/3}\cdot\frac{x-1/3}{2/3}=\frac{x+1}{2}\cdot\frac{3x+1}{4}\cdot\frac{3x-1}{2}=\frac{(9x^2-1)(x+1)}{16}=\frac{9x^3+9x^2-x-1}{16}\ //$$

$$P(x)=\frac{1}{16}\left[\sin(-1)(-9x^3+9x^2+x-1)+\sin(-1/3)(27x^3-9x^2-27x+9)+\sin(1/3)(-27x^3-9x^2+27x+9)+\sin(1)(9x^3+9x^2-x-1)\right]$$

($P(x)$: Exact)

Approx, rounding errors.

$$P(x)=-0.157627x^3+0x^2+0.999098x+0$$

→ swap to t's...

$$P(t)=-0.157627t^3+0.999098t$$

Remark: This is very Close to the Taylor Polynomial.

$$-0.166t^3+1t$$

Lagrange is more accurate, Taylor is more efficent.

Problem 2

Use Gram Schmidt $u_1=1$ $u_2=x$ $u_3=x^2$ to compute 3 polynomials which are orthonormal on $[0,1]$ $\rightarrow (f,g)=\int_0^1 f(x)\,g(x)\,dx$

Then find best approx for $x^{1/2}$ in $P_2[0,1]$

$$V_j = u_j - \sum_{k=1}^{j-1} \frac{(u_j,u_k)}{(v_k,v_k)}\cdot v_k$$

1.) $v_1 = u_1 = 1$

2.) $v_2 = u_2 - \frac{(u_2,u_1)}{(v_1,v_1)} = x - \frac{(x,1)}{(1,1)}\cdot 1 = x - \frac{\int_0^1 x\cdot 1\,dx}{\int_0^1 1\cdot 1\,dx} = \frac{x-x^2/2\big|_0^1}{x\big|_0^1} = \frac{x-1/2}{1}$ //

$$v_3 = u_3 - \sum_{k=1}^{2} \frac{(u_3,u_k)}{(v_k,v_k)}\cdot v_k = u_3 - \frac{(u_3,u_1)}{(v_1,u_1)}\cdot v_1 - \frac{(u_3,u_2)}{(v_2,v_2)}\cdot v_2 = x^2 - \int_0^1 x^2\,dx - \frac{\int_0^1 x^2(x-1/2)\,dx}{\int_0^1 (x-1/2)^2\,dx}\cdot(x-1/2)$$

$\int_0^1 x^2\,dx = 1/3$ // $\int_0^1 x^3 - \frac{1}{2}x^2\,dx = \frac{x^4}{4} - \frac{x^3}{6}\Big|_0^1 = \frac{1}{4} - \frac{1}{6} = \frac{3}{12} - \frac{2}{12} = 1/12$ //

$\int_0^1 (x-1/2)^2\,dx = \frac{(x-1/2)^3}{3}\Big|_0^1 = \frac{(1-1/2)^3}{3} - \frac{(0-1/2)^3}{3}$

$= \frac{1}{24} - \frac{-1}{24} = 1/12$ //

$\int_0^1 (x-1/2)^2\,dx$ $\quad u = x-1/2$, $du = dx$

$\frac{(x-1/2)^3}{3}\Big|_0^1$

$\frac{(x-1/2)^3}{3}\Big|_0^1$

$v_3 = x^2 - 1/3 - \frac{1/12}{1/12}(x-1/2) = x^2 - x - 1/3 + 1/2 = x^2 - x + 1/6$ //

$v_1 = 1$, $v_2 = x - 1/2$, $v_3 = x^2 - x - 1/3$.

$w_1 = \frac{v_1}{\|v_1\|} = \frac{v_1}{(v_1,v_1)^{1/2}} = \frac{1}{1}$ //

$w_2 = \frac{v_2}{(v_2,v_2)^{1/2}} = \frac{x-1/2}{\sqrt{1/12}}$ //

$w_3 = \frac{x^2-x+1/6}{\left(\int_0^1 (x^2-x+1/6)^2\,dx\right)^{1/2}} = \frac{x^2-x+1/6}{\sqrt{1/180}}$ //

$\int_0^1 x^4 - x^3 + 1/6\,x^2 - x^3 + x^2 - 1/6\,x + 1/6\,x^2 - 1/6\,x + 1/36\,dx = \int_0^1 x^4 - 2x^3 + \frac{4}{3}x^2 - 1/3\,x + 1/36 = \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}$

($\frac{1}{3}x^2 + 1x^2$)

↳ 0,1 is easy to int.

$$P_F = \sum_{j=1}^{n} (f,w_j)\cdot w_j = (f,w_1)\cdot w_1 + (f,w_2)\cdot w_2 + (f,w_3)\cdot w_3$$

$= \int_0^1 x^{1/2}\cdot 1\,dx\cdot 1 + \int_0^1 x^{1/2}\left(\frac{x-1/2}{\sqrt{1/12}}\right)dx\cdot\frac{x-1/2}{1/\sqrt{12}} + \int_0^1 x^{1/2}\left(\frac{x^2-x+1/6}{1/\sqrt{180}}\right)dx\left(\frac{x^2-x+1/6}{1/\sqrt{180}}\right)$

$\frac{2}{3}\cdot 1\cdot 1 = \frac{2}{3}$ // $\int_0^1 x^{3/2}\,dx - 1/2\int_0^1 x^{1/2}\,dx$ $\quad \frac{x^{3/2}}{3/2}$

$\frac{x^{5/2}}{5/2} = \frac{2}{5} - \frac{1}{2}\cdot\frac{2}{3}$

$= \frac{2}{5} - \frac{1}{3} = \frac{6}{15} - \frac{5}{15} = 1/15$

$\frac{1}{1/\sqrt{12}}\cdot\frac{1}{15}\left(\frac{x-1/2}{1/\sqrt{180}}\right)$ //

$= \frac{12}{15}(x-1/2)$ //

$180\left(\int_0^1 x^{5/2} - x^{3/2} + 1/6\,x^{1/2}\,dx\right)$

$\frac{x^{7/2}}{7/2} - \frac{x^{5/2}}{5/2} + \frac{1}{6}\frac{x^{3/2}}{3/2}$

$\frac{2}{7} - \frac{2}{5} + \frac{1}{7}$

$\frac{-1}{35}\cdot 180 = \left(x^2 - x + 1/6\right)$

$$P_2(x) = \frac{2}{3} + \frac{12}{15}(x-1/2) - \frac{180}{315}(x^2 - x + 1/6)$$

$$P(x) = \frac{6}{35} + \frac{48}{35}x - \frac{4}{7}x^2$$

Problem 3

Fine the quadratic polynomial that satisfies $P(0) = f(0)$, $P(2) = f(2)$ $P'(2) = f'(2)$

$P_2(x) = a + bx + cx^2$

$P'(x) = b + 2cx$

① $P(0) = a = f(0)$ //

② $P(2) = a + 2b + 4c = f(2)$

③ $P'(2) = b + 2 \cdot c2 = b + 4c = f'(2)$

3 unknowns, 3 equations

$2b + 4c = f(2) - f(0)$

$-(b + 4c = f'(2))$

$b = f(2) - f(0) - f'(2)$ //

at $2b + 4c = f(2)$

$f(0) + 2f(2) - 2f(0) - 2f'(2) + 4c = f(2)$

$4c = \underline{f(2)} - \underline{f(0)} - \underline{2f(2)} + \underline{2f(0)} + 2f'(2)$

$4c = -f(2) + f(0) + 2f'(2)$ //

$c = -1/4 f(2) + 1/4 f(0) + 1/2 f'(2)$

$P(x) = f(0) + (f(2) - f(0) - f'(2))x + (-\frac{1}{4} f(2) + 1/4 f(0) + 1/2 f'(2))x^2$